End-of-Year Internship Report

AI and Data Technologies (IATD-SI)

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# 3D Human Motion Generation and Understanding Through Multilingual Text Descriptions

*Alignment, regularisation, and the generation–discrimination trade-off in MotionCLIP*

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# Abstract

Text-to-motion generation is almost exclusively driven by English. This report documents five weeks of research on extending *MotionCLIP*, a transformer autoencoder that compresses a 60-frame human-motion clip into a single 512-dimensional latent, to condition on text in seven languages (English, French, Spanish, Portuguese, Russian, Chinese, Arabic). We substitute the frozen CLIP text tower with multilingual M-CLIP and, as an alternative, a bank of seven per-language BERT encoders, and we train under a variational objective.

Four findings structure the work. First, in a single-latent motion VAE it is *sampling*, not the strength of the KL penalty, that collapses reconstruction: four interventions spanning two orders of magnitude of $\lambda_{\mathrm{KL}}$ move the reconstruction error by 0.002. A first-order argument explains why a single latent cannot absorb the injected noise, and a deterministic-then-stochastic schedule recovers a working variational model. Second, the cosine alignment objective inherited from MotionCLIP produces *no* per-sample text–motion alignment at any weight, raising its weight 100× makes the collapse worse, while a contrastive objective produces a real, monotonically growing alignment gap across all seven languages. Third, alignment quality and generation quality *dissociate*: a representation can be highly discriminative and poorly decodable at once. Fourth, generation amplitude decays monotonically with training under mean-squared reconstruction, driving the decoder toward a near-static mean pose. A controlled English/Latin/FAR/ALL comparison shows that added languages divide capacity rather than contributing complementary information, and that a combined contrastive-plus-cosine loss restores generation diversity at essentially unchanged FID.

The overarching limitation, and the report's forward proposal, is architectural: a single latent is the wrong container for a multilingually conditioned motion sequence. We argue for a sequence of (discrete) latents and name MoMask with M-CLIP conditioning as the natural next test.

# Résumé

La génération de mouvement à partir de texte est presque exclusivement pilotée par l'anglais. Ce rapport présente cinq semaines de recherche sur l'extension de *MotionCLIP*, un autoencodeur transformeur qui compresse un clip de mouvement de 60 images en un unique vecteur latent de dimension 512, afin de le conditionner sur du texte en sept langues (anglais, français, espagnol, portugais, russe, chinois, arabe). Nous remplaçons l'encodeur de texte CLIP, gelé, par M-CLIP multilingue et, en variante, par un banc de sept encodeurs BERT spécifiques à chaque langue, et nous entraînons le modèle sous un objectif variationnel.

Quatre résultats structurent ce travail. Premièrement, dans un VAE de mouvement à latent unique, c'est *l'échantillonnage*, et non la force de la pénalité KL, qui détruit la reconstruction : quatre interventions couvrant deux ordres de grandeur de $\lambda_{\mathrm{KL}}$ ne déplacent l'erreur de reconstruction que de 0,002. Un argument au premier ordre explique pourquoi un latent unique ne peut absorber le bruit injecté, et une stratégie déterministe-puis-stochastique restaure un modèle variationnel fonctionnel. Deuxièmement, l'objectif cosinus hérité de MotionCLIP ne produit *aucun* alignement texte–mouvement par échantillon, quel que soit son poids ; un objectif contrastif produit, lui, un écart d'alignement réel et croissant dans les sept langues. Troisièmement, la qualité d'alignement et la qualité de génération se *dissocient*. Quatrièmement, l'amplitude du mouvement généré décroît de façon monotone au cours de l'entraînement, l'objectif de reconstruction poussant le décodeur vers une pose moyenne quasi statique.

La limite principale, et la proposition prospective du rapport, est architecturale : un latent unique est un mauvais conteneur pour une séquence de mouvement conditionnée de façon multilingue. Nous plaidons pour une séquence de latents (discrets), avec MoMask conditionné par M-CLIP comme prochaine expérience naturelle.

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# List of abbreviations

| | |
|---|---|
| **CE** | Cross-entropy; here the InfoNCE contrastive loss over in-batch negatives |
| **CLIP** | Contrastive Language–Image Pre-training |
| **FID** | Fréchet Inception Distance (motion-feature version) |
| **GELU** | Gaussian Error Linear Unit activation |
| **KL** | Kullback–Leibler divergence |
| **M-CLIP** | Multilingual CLIP text encoder (XLM-RoBERTa-Large / ViT-B-32) |
| **MM-Dist** | Multimodal distance (text-to-generated-motion feature distance) |
| **MSE** | Mean squared error |
| **NLLB** | No Language Left Behind (machine-translation model) |
| **R@*k*** | Retrieval precision at rank $k$ (R-precision) |
| **rc** | Positional reconstruction loss |
| **VAE** | Variational autoencoder |
| **vel** | Velocity (frame-difference) reconstruction loss |
| **FAR** | The language set {en, ru, zh, ar} (typologically far from English) |
| **Latin** | The language set {en, fr, es, pt} (Latin-script, Romance + English) |

CHAPTER 1

# Introduction

## 1.1 Context and host institution

This internship was carried out at **AI Movement**, the International Artificial Intelligence Center of Morocco, a research centre of Mohammed VI Polytechnic University (UM6P) established under the auspices of UNESCO. The centre conducts research across computer vision, natural language processing, and embodied AI. The work reported here sits at the intersection of the first two: generating three-dimensional human motion from natural-language descriptions, with an emphasis on languages other than English.

The project ran for five weeks on a single shared RTX 5090 workstation. The deliverable was twofold: a body of controlled experiments answering a research question posed by the supervisors, and a repository clean enough to be handed over and continued. This report is the handover document. It is written so that a reader can understand every experiment that was run and why, see the numbers, and locate the script that produced each of them.

## 1.2 Problem statement

Text-to-motion generation, synthesising a plausible sequence of 3D body poses from a sentence such as *"a person walks forward and then sits down"*, has advanced rapidly, but almost entirely in English. The dominant benchmark, HumanML3D [7], is English-only, and the strongest models [18, 6] are trained and evaluated exclusively on English captions. A user who describes motion in French, Russian, or Arabic is, in practice, excluded.

Extending these systems to be multilingual raises a question that is not merely engineering. Human languages differ in how they encode *manner* (how a motion is performed), *path* (its trajectory), and *aspect* (its temporal structure). If a multilingual model could exploit these differences, additional languages might supply complementary signal about the motion itself. If instead the languages merely contend for a fixed representational budget, adding them would degrade the shared space. Which of these occurs is an empirical question, and it has not been answered under controlled conditions.

## 1.3 Research question

The concrete question this internship set out to answer, posed by the supervisors, is:

> *Do independently trained per-language text projections bring several languages into a common motion latent space, and does multilingual conditioning help or harm the underlying motion representation, once capacity and compute are matched?*

Answering this required a comparison that, to our knowledge, has not been run: English-only versus multilingual conditioning, *matched for model capacity and total gradient steps*, evaluated on a common English test set so that the numbers are comparable. It also required localising *where* any cross-lingual degradation originates, in the frozen text encoder, or in the trained motion latent, by measuring text-space alignment and motion-latent alignment separately. A recurring theme of the report is that this second measurement, done carelessly, gives exactly the wrong answer.

## 1.4 Contributions

1. **Sampling, not KL regularisation, is what breaks reconstruction** in a single-latent motion VAE. We show this with four controlled interventions spanning two orders of magnitude of $\lambda_{\mathrm{KL}}$, which together move reconstruction error by 0.002 (Section 4.3), and give a first-order argument for why a single latent cannot absorb the noise (Section 4.3.1).

2. **A working variational model** obtained by deterministic-then-stochastic training, restoring reconstruction with sampling active (Section 4.4).

3. **The cosine alignment objective produces no per-sample alignment at any weight**; a contrastive objective fixes it across seven languages, with a stable per-language ordering (Sections 4.6–4.7).

4. **Alignment and generation quality dissociate** (Section 4.8), and **generation amplitude decays monotonically with training** under MSE reconstruction, collapsing the decoder toward a near-static mean pose (Section 4.9).

Two negative results are reported in full because they are informative: a noise-robustness fine-tune that improves an auxiliary metric but degrades generation (Section 4.5), and the first per-language-encoder comparison, which was measured under an objective later shown to be uninformative (Section 4.2).

## 1.5 Report outline

Chapter 2 reviews motion representation, text-to-motion generation, variational autoencoders, and multilingual representation learning. Chapter 3 describes the architecture, the multilingual pipeline, the loss, and, critically, the evaluation protocol and its two non-comparable evaluators. Chapter 4 is the core: it presents the experiments in the narrative order in which each result motivated the next. Chapter 5 states the threats to validity bluntly and develops the forward proposal. Chapter 6 concludes. Appendix A gives implementation details, Appendix B is a complete index of every run and the script that produced it, and Appendix C collects additional tables and figures.

CHAPTER 2

# Background and Related Work

## 2.1 Human motion representation

We use the HumanML3D dataset [7], which pairs 3D human motions with English captions. Each motion is a sequence of frames, and each frame is encoded as a **263-dimensional** redundant pose vector: root angular and linear velocity, root height, local joint positions, joint velocities, and foot-contact indicators, derived from the SMPL body model [11]. This report works with fixed-length clips of **60 frames**, so a single clip is a $60 \times 263 = 15{,}780$-number tensor. After filtering, the corpus provides **22,326** training clips (966 discarded as too short, 92 as too long) and **1,400** validation clips.

## 2.2 Text-to-motion generation

Three systems frame where this work sits. **MotionCLIP** [17] is a transformer encoder–decoder that aligns a single motion latent to the frozen CLIP embedding space; it is a weak generator but an unusually simple and transparent one, which is why it is the base model here. **MDM** [18] is a diffusion model over the raw motion sequence, and **MoMask** [6] is a residual-vector-quantised masked transformer that produces a *sequence of discrete tokens*. MoMask and MDM are far stronger generators; MoMask's token-sequence design is precisely the architecture our closing argument points towards.

> **A calibration the reader should keep in mind throughout.** Published HumanML3D FID for strong models is in the range 0.05–0.5 [6]. Every FID in this report is one to two orders of magnitude larger. MotionCLIP is a *weak* generator, and the comparisons here are *relative*, not absolute.

## 2.3 Variational autoencoders

A VAE [9] encodes an input to a distribution $q(z \mid x) = \mathcal{N}(\mu, \sigma^2)$ rather than a point, samples $z = \mu + \sigma \odot \varepsilon$ via the reparameterisation trick, and regularises the posterior toward a unit Gaussian with a KL term. The sampling step is what makes the latent space smooth and generative; it is also, we will show, what a single-latent motion decoder cannot tolerate. **ACTOR** [14] is the action-conditioned transformer-VAE precedent that survives sampling, a fact we explain in Section 4.3.1 by the far lower information it passes through the same bottleneck.

## 2.4 Multilingual representation learning

**CLIP** [15] aligns images and English text with a contrastive (InfoNCE [19]) objective. **M-CLIP** [3] replaces CLIP's text tower with XLM-RoBERTa-Large [4], trained so that translations of a caption map near the original English CLIP embedding; it is multilingual out of the box. To obtain non-English captions we machine-translate HumanML3D with **NLLB** [13] into six target languages. The operative property of a contrastive objective, and the one the cosine objective lacks, is the presence of *negatives*: it constrains matched pairs to rank above mismatched pairs, not merely to be similar.

## 2.5 Multilingual motion generation

Multilingual text-to-motion is an emerging direction; recent work explores bilingual extensions of HumanML3D and cross-lingual alignment modules that map non-English text into an English-trained motion space [21], most directly the BiHumanML3D benchmark, which pairs a bilingual re-annotation of HumanML3D with a cross-lingual alignment module. The common assumption is that multilingual signal is, at worst, neutral for the underlying motion representation. This report tests that assumption directly and, under matched capacity and compute, finds it does not hold: added languages divide a fixed representational budget rather than enriching it (Section 4.11).

CHAPTER 3

# Methodology

## 3.1 Base architecture

MotionCLIP is a transformer encoder–decoder (8 layers, GELU activations) [17, 20]. The encoder reads a 60-frame clip and, through a learned query token, compresses it into a **single 512-dimensional latent**; the decoder reconstructs all 60 frames from that one vector. This is a 30× compression through a single point, and it is the fact that drives most of this report: one vector must carry an entire clip, so any noise injected into it damages the whole sequence at once (Section 4.3.1), and any objective that under-constrains it collapses generation (Sections 4.8–4.9).

## 3.2 Multilingual pipeline

### 3.2.1 Text encoder substitution

The original CLIP text tower is replaced by **M-CLIP** (XLM-RoBERTa-Large / ViT-B-32), kept *frozen*, with a single trainable 768 → 512 linear projection into the motion latent space.

### 3.2.2 Caption translation

HumanML3D captions are machine-translated with NLLB into six languages, `fr es pt ru zh ar`, giving seven in total with English. During training, each clip is presented in a language sampled per clip per epoch, so that at a fixed number of gradient steps a clip is seen in any given language roughly $1/N$ of the time under $N$-language training. This exposure imbalance is a confound we return to in Section 5.2.

### 3.2.3 Per-language encoder bank

As an alternative to the shared M-CLIP tower, we assemble a **bank of seven frozen monolingual encoders**, each with its own trainable 768 → 512 projection: `bert-base-uncased` [5], `camembert-base` [12], Spanish BETO [2], Portuguese BERTimbau [16], Russian RuBERT [10], `bert-base-chinese`, and Arabic AraBERT [1]. This is **854M** frozen parameters and **2.76M** trainable projection parameters. The supervisors' question, whether these *independently* trained projections land in a common space, is what motivates this design.

## 3.3 Variational formulation

The training loss is a weighted sum of five terms:

$$L \;=\; 100 \cdot \mathrm{RC} \;+\; 100 \cdot \mathrm{VEL} \;+\; \lambda_{\cos} \cdot \cos \;+\; \lambda_{\mathrm{CE}} \cdot \mathrm{CE} \;+\; \lambda_{\mathrm{KL}} \cdot \mathrm{KL}. \tag{3.1}$$

Table 3.1 lists the terms. The 100:1 weighting of reconstruction to cosine is inherited from MotionCLIP [17] and is *not* tuned in this work; we state this explicitly because Section 4.6 shows the cosine term to be the wrong objective regardless of its weight.

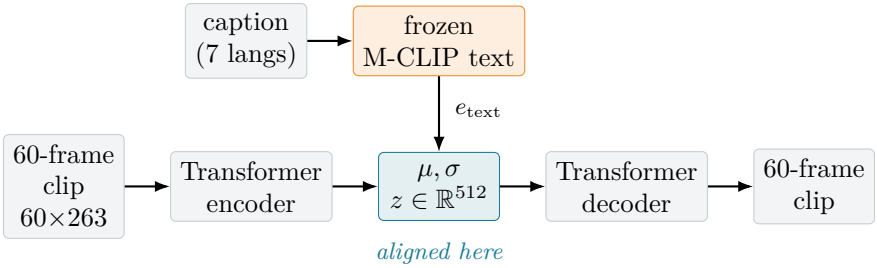



**Figure 3.1.** The system. A 60-frame clip is compressed by the encoder into a single 512-dim latent $z$ and reconstructed by the decoder. A frozen multilingual text encoder (M-CLIP, or a per-language bank) produces $e_{\text{text}}$, which is aligned to the motion latent $\mu$ during training. At generation time the decoder is fed $e_{\text{text}}$ in place of $\mu$: a substitution that the base loss never optimises (Section 4.10).

**Table 3.1.** Loss terms in Equation 3.1. Default weights: $\lambda_{\text{rc}} = \lambda_{\text{vel}} = 100$, $\lambda_{\text{KL}} = 10^{-3}$; $\lambda_{\text{cos}}$ and $\lambda_{\text{CE}}$ vary by experiment.

| Term | Symbol | Definition |
|---|---|---|
| Reconstruction | RC | positional MSE, $\|\hat{x} - x\|^2$ |
| Velocity | VEL | MSE on frame-to-frame differences; suppresses jitter |
| Cosine | cos | mean $(1 - \cos(e_{\text{text}}, \mu))$, matched pairs only |
| Contrastive | CE | InfoNCE over in-batch negatives |
| KL | KL | $-\frac{1}{2}\sum(1 + \log\sigma^2 - \mu^2 - \sigma^2)$ |

## 3.4 Alignment objectives

### 3.4.1 Cosine similarity

The cosine term pulls each matched text embedding and motion latent together. Crucially it involves *matched pairs only*: it contains no negatives, so it never pushes mismatched pairs apart.

### 3.4.2 Contrastive cross-entropy

The contrastive term is InfoNCE [19] over in-batch negatives: for each motion it raises the similarity of its own caption above that of every other caption in the batch. The presence or absence of negatives is the operative difference between the two objectives, and Section 4.6 shows it to be decisive.

## 3.5 Evaluation protocol

> **Two evaluators are used and their numbers are never comparable. Evaluator A** is the Guo/MoMask evaluator on the *official* HumanML3D test split; it reports FID ≈ 16 and is used only for the M-CLIP baseline table (Section 4.1). **Evaluator B** (`eval_guo_vae_compare_beta05.py`) runs on the 1,400-clip validation pool and reports FID ≈ 56–120; it is used for all VAE comparisons. A number from A and a number from B must never appear in the same comparison.

### 3.5.1 Generation metrics

Both evaluators report **FID** [8] (lower is better), **R-precision** (R@1/2/3/10, retrieval of the correct caption), **MM-Dist** (text-to-motion feature distance, lower is better), and **Diversity**, each over **20 replications**. One caveat must be stated up front: **R-precision is English-only by construction**. The Guo evaluator's text tower is a GloVe BiGRU trained on English

captions, so it can score retrieval only in English; per-language quality is therefore assessed by FID alone.

### 3.5.2 Latent alignment measurement

To measure whether text and motion are aligned *per sample*, we compute cosine similarity for **matched** versus **mismatched** text–motion pairs over 500 clips with parallel captions in all seven languages, and report the **gap** between them. As Section 4.6 shows, the *mean* matched cosine alone is actively misleading: it can be near 1.0 while the gap, the quantity that actually reflects alignment, is zero.

### 3.5.3 Motion amplitude

We introduce a diagnostic used throughout Section 4.9: the **mean absolute frame-to-frame difference** of a generated motion (its overall movement), together with the difference between motions generated for different prompts (a measure of how much the model conditions on text).

CHAPTER 4

# Experiments and Results

This chapter is written as a sequence in which each result motivates the next. It begins with a multilingual baseline that works, turns to a per-language variant that appears to fail, and then, through the KL series, discovers that the objective under which that failure was measured is itself uninformative. The remainder builds the correct alignment measurement, exposes the generation–discrimination trade-off and the amplitude collapse, and closes with the controlled language comparison that was the internship's stated question.

## 4.1 Multilingual baseline

We first establish that multilingual conditioning works at all. Table 4.1 reports the M-CLIP variants under **Evaluator A** (official split, 20 replications), grouped by language set: *Latin* = {en, fr, es, pt}, *FAR* = {en, ru, zh, ar}.

**Table 4.1.** Multilingual baseline. **Evaluator A**, official HumanML3D test split, 20 replications. Best model value in each column in bold.

| Model | FID↓ | Diversity | R@1 | R@2 | R@3 | MM-Dist↓ |
|---|---|---|---|---|---|---|
| Ground truth | 0.010 | 9.465 | 0.527 | 0.714 | 0.813 | 2.924 |
| M-CLIP EN | 18.013 | 7.215 | 0.128 | 0.235 | 0.327 | 6.002 |
| M-CLIP Latin | **16.140** | 6.697 | **0.136** | **0.244** | **0.344** | **5.757** |
| M-CLIP FAR | 18.046 | 7.002 | 0.111 | 0.215 | 0.304 | 6.003 |
| M-CLIP ALL | 17.829 | 7.092 | 0.119 | 0.220 | 0.310 | 6.057 |

The Latin set gives the best FID (16.14) and the best retrieval, slightly ahead of English-only. Multilingual conditioning is not merely tolerated; on this split it helps. This is the encouraging result that motivated the rest of the internship.

## 4.2 Per-language encoders, first attempt

We next replaced the shared M-CLIP tower with the per-language bank (Latin subset, four frozen backbones). Under **Evaluator A** this appeared to fail badly: FID 24.55 versus 16.14, R@1 0.079 versus 0.136, and diversity fell to ∼4.0. Reconstruction, however, was preserved (RC = 0.086 versus 0.083), so the decoder was intact and *alignment* was the casualty.

> **Critical caveat, stated up front.** This comparison was measured under the *cosine* objective, which Section 4.6 shows produces no per-sample alignment in *any* condition. The comparison is therefore uninformative about the per-language encoders and is re-run under a contrastive objective in Section 4.10, where the verdict reverses.

This apparent failure is the origin of the supervisors' question: do the independently trained projections share a space? Answering it properly is what forced the alignment measurement of Section 4.6.

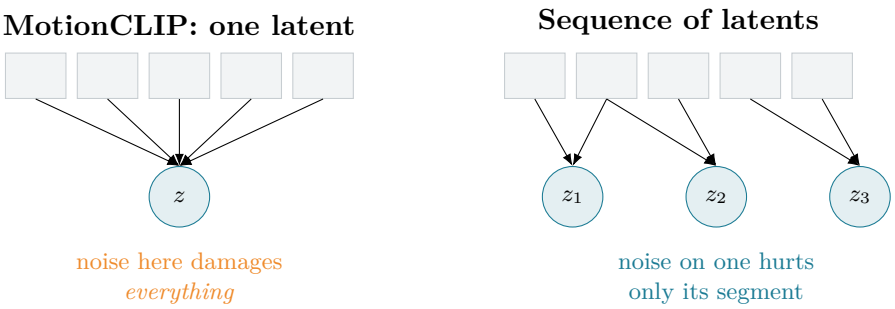



**Figure 4.1.** Why a single latent cannot absorb sampling noise. Left: all 60 frames depend on one $z$, so injected noise damages the whole clip (Equation 4.1 sums over every frame). Right: with a latent per frame-span, most Jacobian blocks vanish and the penalty per latent falls by roughly $K$. This is the architectural argument developed in Section 5.3.

## 4.3 The KL series: sampling, not regularisation

Moving to a variational model, reconstruction immediately collapsed. The natural suspect was the KL penalty. It is not. Table 4.2 varies the KL treatment across four interventions spanning two orders of magnitude of $\lambda_{\text{KL}}$.

**Table 4.2.** The KL series (sampling active). Reconstruction loss under four interventions. Total spread 0.002.

| Configuration | RC↓ |
|---|---|
| Deterministic autoencoder (no sampling) | **0.083** |
| VAE, $\lambda_{\text{KL}} = 10^{-5}$ | 0.524 |
| VAE, $\lambda_{\text{KL}} = 10^{-3}$ (100× stronger) | 0.526 |
| VAE, cosine removed (ACTOR-style) | 0.524 |
| VAE, $\lambda_{\text{KL}}$ annealed over 30 epochs | 0.525 |

The deterministic autoencoder reconstructs at 0.083; the moment sampling is switched on, reconstruction jumps to ∼0.52 and *stays there*, whether the KL weight is $10^{-5}$ or $10^{-3}$, whether the cosine term is present or removed, and whether the penalty is annealed. Four interventions, total spread **0.002**. **Sampling is the binding constraint, not the KL weight.**

### 4.3.1 Why the bottleneck cannot absorb noise

The reason is structural. Expanding the reconstruction error to first order in the sampling noise around $\mu$, with $z = \mu + \varepsilon$ and per-frame decoder Jacobians $J_t = \partial \hat{x}_t / \partial z$:

$$\mathbb{E}_\varepsilon \|\hat{x} - x\|^2 \;\approx\; \|\hat{x}(\mu) - x\|^2 \;+\; \sum_{t=1}^{60} \operatorname{tr}\!\left(J_t \, \Sigma \, J_t^\top\right). \tag{4.1}$$

The damage term sums over **all 60 frames**, because every $J_t$ depends on the *same* $z$. There is nowhere for the noise to hide: one vector feeds the entire clip. With $K$ latents each covering a disjoint span of frames, most $\partial \hat{x}_t / \partial z_k = 0$, the sum becomes block-diagonal, and the penalty per latent falls by roughly a factor of $K$ (Figure 4.1).

This also explains why ACTOR [14] survives sampling where MotionCLIP does not: ACTOR conditions on a discrete action class, so it passes far less information through the same bottleneck and has a lower-gain decoder, smaller $J_t$, smaller damage term.

## 4.4 Deterministic reconstruction, and a working VAE

If sampling is the problem, the fix is to reconstruct from $\mu$ while keeping the KL term. Doing so recovers reconstruction from 0.52 to **0.16**. The KL benchmark under **Evaluator B** (Table 4.3) then shows that KL genuinely helps generation, and that annealing hurts in both schedules tried.

**Table 4.3.** KL benchmark. **Evaluator B**, 1,400-clip validation pool, 20 replications. KL helps; annealing hurts.

| Model | FID↓ |
|---|---|
| $\lambda_{\mathrm{KL}} = 0$ | 97.633 |
| $\lambda_{\mathrm{KL}} = 10^{-3}$ | **55.864** |
| $\lambda_{\mathrm{KL}} = 10^{-3}$ + anneal over 30 epochs | 105.683 |
| $\lambda_{\mathrm{KL}} = 10^{-3}$ + anneal over 5 epochs | 100.893 |

Annealing never had a fair test. In `reparameterize()`, sampling activates as soon as $\lambda_{\mathrm{KL}} \neq 0$, so every annealed run trained with *full-magnitude noise from epoch 1* while only the *penalty* was ramped. Separating the two variables gives Table 4.4: train deterministically to convergence, then enable sampling.

**Table 4.4.** Deterministic-then-stochastic training. Sampling is enabled only after 300 deterministic epochs.

| Configuration | Sampling | RC↓ |
|---|---|---|
| Sampling from initialisation | on | ∼0.52 |
| Deterministic, 300 epochs | off | 0.093 |
| + 10 epochs fine-tune | **on** | **0.086** |

**Caveat that must not be dropped.** Two variables changed between the last two rows: sampling was enabled *and* 10 further epochs were run. The defensible claim is that *sampling no longer destroys reconstruction*, not that it improves it. The missing control, 10 further *deterministic* epochs from the same checkpoint, is an open thread (Appendix B). This number is also a training-loss average; held-out confirmation across noise levels is pending because `eval_noise_val100.py` loads hardcoded checkpoints and ignored the `--ckpt` override.

## 4.5 Noise robustness: a deliberately closed branch

A tempting way to make the latent tolerate sampling is to fine-tune under injected noise $z = \mu + \beta\varepsilon$. Reconstruction degrades smoothly with $\beta$ (Table 4.5), and fine-tuning at $\beta = 0.10$ gave a **+25.5%** reconstruction improvement: **92%** of clips improved, bootstrap 95% CI $[0.227, 0.373]$. Statistically, this is solid.

**Table 4.5.** Reconstruction under injected noise $z = \mu + \beta\varepsilon$.

| $\beta$ | 0 | 0.1 | 0.25 | 0.5 | 1.0 |
|---|---|---|---|---|---|
| RC | 0.0805 | 0.0922 | 0.1621 | 0.3857 | 0.6529 |

Downstream, it fails. Table 4.6 shows the $\beta = 0.10$ fine-tune *degrading* every generation metric. Teacher distillation and encoder freezing were also tried and did not recover it. The branch was closed.

**Table 4.6.** Downstream effect of the $\beta = 0.10$ fine-tune. **Evaluator B**. The auxiliary gain does not transfer.

| | R@1 | R@3 | FID↓ | MM-Dist↓ |
|---|---|---|---|---|
| original | **0.062** | **0.104** | **57.317** | **7.274** |
| $\beta = 0.10$ fine-tuned | 0.031 | 0.073 | 86.774 | 9.165 |

**The negative transfer is the point.** An improvement on an auxiliary metric (reconstruction under noise) does not reach generation by default; it can actively harm it. We report the branch in full for exactly this reason.

## 4.6 Alignment is not what mean cosine measures

We now return to the supervisors' question and measure alignment properly. The multilingual VAE `vae-multilingual-mclip` (20 epochs) trains cleanly: RC = 0.1550, VEL = 0.0594, KL = 76.77, text_cosine = 0.3127. Its per-language FID on the multilingual pool (**not** the official split; overall R@1 0.042, R@3 0.104, FID 120.031, MM-Dist 9.946) is uniform across languages (Table 4.7), and cross-language generation drift from the English version is tiny (0.010–0.017).

**Table 4.7.** Per-language FID, multilingual pool (**Evaluator B**, not the official split). Uniform across languages.

| ZH | PT | RU | FR | AR | EN | ES |
|---|---|---|---|---|---|---|
| 99.777 | 106.077 | 108.627 | 108.665 | 112.151 | 114.813 | 124.894 |

Text-space alignment among M-CLIP embeddings (100 shared clips) is very high: EN–FR 0.9721, EN–ES 0.9750, EN–PT 0.9744, EN–RU 0.9823, EN–ZH 0.9762, EN–AR 0.9700; mean EN↔non-EN **0.9750**, full pairwise range 0.963–0.982.

> **Caveat.** This last measurement is of *M-CLIP*, a frozen pretrained model. It says the text embeddings are already aligned across languages; it says *nothing* about what the VAE learned about motion.

The measurement that matters is matched versus mismatched similarity over 500 clips with identical motions and parallel captions (Table 4.8). Chance R@1 over 500 candidates is 0.002.

**Table 4.8.** Matched vs mismatched cosine, 500 clips, parallel captions (English). The gap, not the mean, reflects alignment.

| Objective | matched | mismatched | **gap** | R@1 |
|---|---|---|---|---|
| cosine, $\lambda = 1$ | 0.7284 | 0.7282 | 0.0002 | 0.008 |
| cosine, $\lambda = 100$ | 0.9777 | 0.9770 | 0.0007 | 0.012 |
| contrastive CE | **0.0459** | 0.0273 | **0.0186** | **0.068** |

Under the cosine objective the gap is essentially zero, and R@1 sits at chance. Raising the cosine weight 100× makes it *worse*: the per-clip standard deviation *fell* from 0.073 to 0.005, i.e. the representation collapsed harder. The English-only control gives gap 0.0000.

**Why:** the cosine objective sums over matched pairs only. Setting every $\mu_i$ to the mean text direction is a global optimum, the loss falls while matched and mismatched stay identical. No weight fixes a missing term. Note that the mean cosine *falls* from 0.978 to 0.046 across the fix while the model improves by an order of magnitude in R@1; **we therefore recommend**

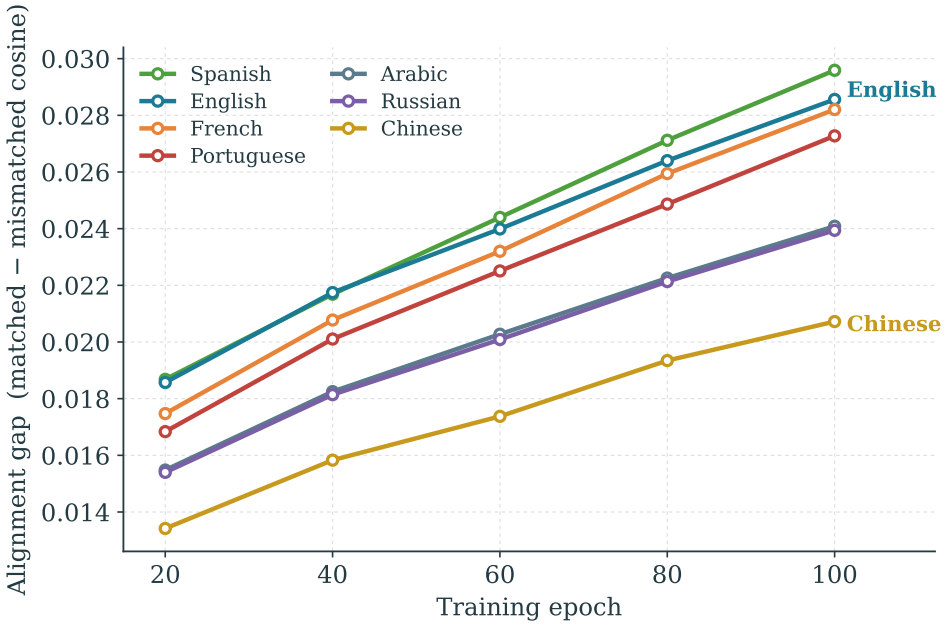



**Figure 4.2.** Alignment gap (matched − mismatched cosine) versus training epoch, one line per language, contrastive CE. The gap grows monotonically in every language with no plateau. The language ordering es ≈ en > fr > pt > ar ≈ ru > zh is *identical at every checkpoint*: structure, not noise. Chinese trails English by 28% at epoch 100.

**against mean cosine as a diagnostic for this model class**, and use the matched-minus-mismatched gap throughout.

## 4.7 Contrastive alignment across languages

The contrastive objective produces a real, growing gap in every language. Table 4.9 tracks it over training; Figure 4.2 plots it.

**Table 4.9.** Per-language alignment gap under contrastive CE, over training. Monotonic in every language; ordering identical at every checkpoint.

| epoch | en | fr | es | pt | ru | zh | ar |
|---|---|---|---|---|---|---|---|
| 20 | 0.0186 | 0.0175 | 0.0187 | 0.0168 | 0.0154 | 0.0134 | 0.0155 |
| 40 | 0.0217 | 0.0208 | 0.0217 | 0.0201 | 0.0181 | 0.0158 | 0.0182 |
| 60 | 0.0240 | 0.0232 | 0.0244 | 0.0225 | 0.0201 | 0.0174 | 0.0203 |
| 80 | 0.0264 | 0.0259 | 0.0271 | 0.0249 | 0.0221 | 0.0193 | 0.0223 |
| 100 | **0.0286** | 0.0282 | **0.0296** | 0.0273 | 0.0239 | 0.0207 | 0.0241 |

The ordering es ≈ en > fr > pt > ar ≈ ru > zh holds at every checkpoint, and the longer 300-epoch run shows no plateau (Appendix C, Figure C.1). Retrieval at 300 epochs, batch 128, is R@1 0.068, R@3 0.174, R@10 0.340 against chance 0.002/0.006/0.020. Alignment is real and well above chance.

## 4.8 The generation–discrimination trade-off

The contrastive objective that fixes alignment *costs* generation quality. Table 4.10 places the two objectives side by side.

**Table 4.10.** The generation–discrimination trade-off. **Evaluator B**. Contrastive CE buys alignment and retrieval at the cost of FID.

| Objective | gap (en) | R@10 | FID↓ | Diversity |
|---|---|---|---|---|
| cosine | 0.0002 | 0.024 | **55.864** | – |
| contrastive CE, 300 ep | **0.0286** | **0.340** | 72.902 | 0.922 |
| real motion | – | – | 0.000 | 9.675 |

The CE-300 row in full: R@1 0.053, R@2 0.098, R@3 0.143, MM-Dist 8.485. At epoch 50 it is R@1 0.069, R@3 0.177, FID 100.781, Diversity 0.750, so the diversity collapse is present *early* and is not overtraining. Rescaling text embeddings to the motion-latent norm (20.413 vs 9.232) changed FID by only 2.1 (72.902 → 70.786) and Diversity by 0.03: **it is not a scale mismatch**.

**Why they dissociate:** InfoNCE constrains only the *ranking* of matched against mismatched pairs, a scale-free, direction-only property. It never requires the latent to retain enough information to *reconstruct* a distinct motion, only enough to be *distinguishable*. A representation can be highly discriminative and poorly decodable at the same time.

## 4.9 Motion amplitude collapse

The most consequential generation failure is that motions collapse toward a near-static pose. Using the amplitude diagnostic of Section 3.5, Table 4.11 shows amplitude *falling* as models train longer and better on every other metric.

**Table 4.11.** Motion amplitude (mean absolute frame-to-frame difference) and between-prompt difference (walk vs jump). Better-trained models move less.

| Model | epochs | amplitude | walk-vs-jump | ratio |
|---|---|---|---|---|
| vae-en-mu-kl1e-3 (cosine) | 20 | **0.0768** | 0.1127 | 1.5 |
| vel10-test | 60 | 0.0145 | 0.0526 | 3.6 |
| ce150-combined | 150 | 0.0108 | 0.0409 | 3.8 |
| ce150-all | 150 | 0.0063 | – | – |
| vae-ml-sampled-ft | 330 | 0.0044 | 0.0204 | 4.6 |

Three explanations are ruled out, one at a time.

1. **Not a rendering artefact.** The 20-epoch checkpoint renders visibly different poses per prompt through the identical script.
2. **Not a text-embedding scale mismatch.** Rescaling made amplitude *worse* (0.01075 → 0.00607).
3. **Not the velocity weight.** At matched epochs, $\lambda_{\text{vel}} = 10$ gave slightly *less* amplitude than $\lambda_{\text{vel}} = 100$ (0.0145 at 60 ep vs 0.0160 at 50 ep).

What it *is*: monotonic decay within a single run, everything else fixed (Table 4.12, Figure 4.3).

**Table 4.12.** Amplitude decay within a single run, all else fixed.

| epochs | ce150-combined | ce150-all |
|---|---|---|
| 50 | 0.01604 | 0.01626 |
| 100 | 0.01249 | 0.00929 |
| 150 | 0.01075 | 0.00626 |

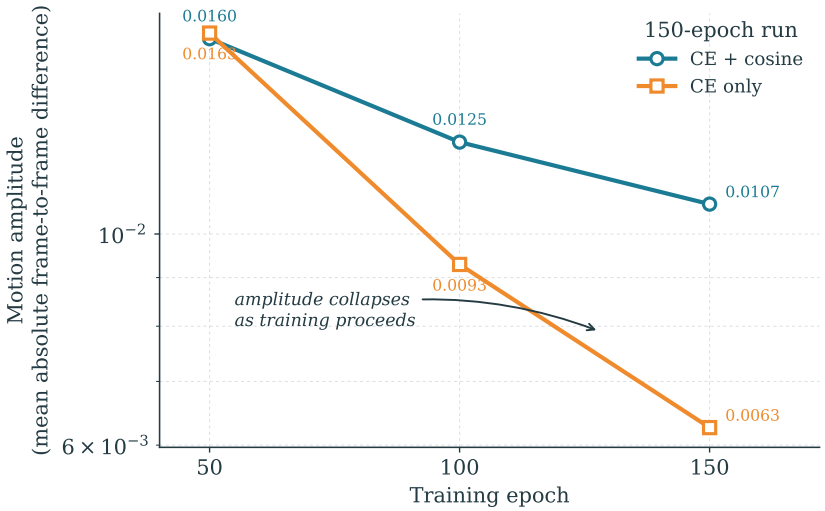



**Figure 4.3.** Motion amplitude versus training epoch for two runs (log $y$-axis). Amplitude decays monotonically as training proceeds. The apparently "better" 20-epoch model of Table 4.11 is simply *less converged*: its high amplitude is undertrained variance, not superior motion.

**Interpretation.** MSE reconstruction drives the decoder toward conditional means. Averaged over pooled motion, that mean is a near-static standing pose. Early checkpoints have not converged there yet, so they still produce variation; the apparently better 20-epoch model is simply less converged. This is visible directly in the generated skeletons (Figure 4.4), which barely move across 60 frames.

We also verified that the *encoder* is healthy: motion–motion cosine over 200 clips spans 0.31–0.98 (mean 0.61), so it produces well-separated latents. The collapse is on the *decoder* side, and only on the text-conditioned path.

## 4.10 Text-conditioned reconstruction

The decoder is trained only on motion latents $\mu$, but at generation time it is fed a *text* embedding $e_{\text{text}}$ (Figure 3.1). That substitution is never optimised. We add a term that trains exactly the path generation uses:

$$L_{\text{text-rc}} \;=\; \|D(e_{\text{text}}) - x\|^2. \tag{4.2}$$

Table 4.13 reports 150-epoch results.

**Table 4.13.** Text-conditioned reconstruction and encoder choice, 150 epochs. "walk-vs-jump" is the between-prompt difference (conditioning strength).

| Run | text_rc | $\lambda_{\text{rc}}$ | encoder | amplitude | walk-vs-jump | ratio |
|---|---|---|---|---|---|---|
| ce150-combined | ✗ | 100 | M-CLIP | 0.0108 | 0.0409 | 3.8 |
| wk-rc20 | ✗ | 20 | M-CLIP | 0.0078 | 0.0376 | 4.8 |
| **wk-textrc** | ✓ | 100 | M-CLIP | 0.0042 | **0.2456** | **58** |
| **wk-textrc-rc20** | ✓ | 20 | M-CLIP | 0.0030 | **0.2933** | **99** |
| wk-perlang | ✗ | 100 | per-lang | **0.0174** | 0.0934 | 5.4 |
| wk-perlang-comb | ✗ | 100 | per-lang | **0.0192** | 0.0972 | 5.1 |

Two findings follow.

**Text conditioning rose 6–7×** (0.041 → 0.293): the decoder now responds strongly to which sentence it is given. But amplitude fell further, to 0.003, the term rewards getting the *pose* right for a clip, not the *trajectory*, since one text embedding must explain all 60 frames.




**Figure 4.4.** Generated skeletons, five prompts (rows) sampled at five time-steps (columns) from a trained model, fixed camera and crop. Within each row the pose barely changes across the 60 frames, and across rows the poses are near-identical, the amplitude collapse of Section 4.9 made visible. Frames rendered from `gen_vae/`.

The result is distinct poses per prompt with almost no movement within each. The natural fix, applying the velocity term to the text-conditioned output as well, to restore trajectory supervision on that path, is proposed but not yet run (Appendix B).

**The per-language encoder bank gives 2–3× more amplitude** than M-CLIP under matched settings (0.017–0.019 vs 0.006–0.011). This *reverses* the apparent verdict of Section 4.2, which was obtained under the broken cosine objective and therefore never measured what it appeared to.

## 4.11 Controlled comparison: EN / Latin / FAR / ALL

The encouraging baseline of Section 4.1, where Latin FID (16.14) beat English-only (18.01), and the English degradation reported just below are not in tension: they are read off different instruments. Section 4.1 measured FID on the *official* HumanML3D split under the cosine objective (Evaluator A) over a longer epoch budget; this section measures retrieval and alignment under contrastive CE at matched compute (Evaluator B). Same phenomenon, different instruments, and it is the comparison in this section that is the controlled one.

Finally, the internship's stated question. All conditions are trained for 150 epochs, batch 128, contrastive CE, $\lambda_{\text{KL}} = 10^{-3}$, M-CLIP text encoder, *identical in every respect except the language set*. Alignment is measured over 500 clips with parallel captions; generation via the Guo evaluator, 20 replications.

**Table 4.14.** Controlled language comparison. All conditions identical except the language set. English retrieval, English alignment gap, cross-lingual spread, and generation. Best value per column in bold.

| Condition | languages | en R@1 | en gap | spread | FID↓ | Diversity |
|---|---|---|---|---|---|---|
| EN | en | **0.610** | **0.0515** | 0.0398 | 67.660 | **1.800** |
| Latin | en fr es pt | 0.410 | 0.0372 | 0.0253 | 95.969 | 0.668 |
| FAR | en ru zh ar | 0.316 | 0.0336 | 0.0096 | **63.924** | 0.760 |
| ALL | all 7 | 0.316 | 0.0334 | 0.0071 | 74.035 | 0.831 |
| ALL + cosine | all 7 | 0.266 | 0.0289 | **0.0066** | 74.479 | 1.459 |
| real motion | – | 0.529 | – | – | 0.000 | ∼9.5 |

**Multilingual conditioning degrades English monotonically.** English retrieval falls 0.610 → 0.410 → 0.316 as languages are added, and diversity drops from 1.800 to 0.73–0.83 for every multilingual condition.[1] This supports the *lossier-shared-space* hypothesis over the richer-supervision hypothesis: added languages induce capacity contention rather than contributing complementary information about manner, path, or aspect.

> **The confound, restated.** At a fixed number of gradient steps, each clip is seen in English roughly $1/N$ as often under $N$-language training. Compute is matched; per-language *exposure* is not. Any English degradation may reflect reduced English exposure rather than multilinguality as such (Section 5.2).

**But the degradation is not general, capacity is divided, not lost.** Table 4.15 shows each condition is best on exactly the languages it trained on.

[1] The two `eval_all.sh` runs report English diversity as 1.800 and 1.798; the evaluator has run-to-run variance of ∼0.002, which we quote as 1.800 (Table 4.14) throughout.

**Table 4.15.** Per-language R@1 by training condition. Each condition is best on its own languages; ALL buys the most uniform coverage at the cost of peak performance. Best value per language in bold.

| condition | en | fr | es | pt | ru | zh | ar |
|---|---|---|---|---|---|---|---|
| EN | **0.610** | 0.182 | 0.188 | 0.198 | 0.118 | 0.066 | 0.084 |
| Latin | 0.410 | **0.388** | **0.430** | **0.438** | 0.150 | 0.100 | 0.086 |
| FAR | 0.316 | 0.150 | 0.172 | 0.144 | **0.288** | **0.238** | **0.260** |
| ALL | 0.316 | 0.272 | 0.294 | 0.298 | 0.248 | 0.172 | 0.184 |

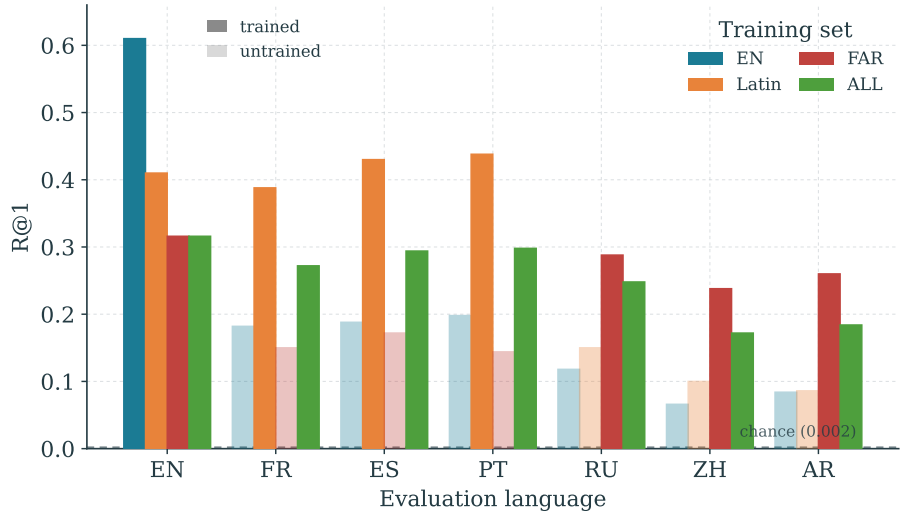



**Figure 4.5.** Per-language R@1 across the four language conditions. Latin lifts fr/es/pt; FAR lifts ru/zh/ar; ALL gives the most uniform coverage. Each bar group peaks on the languages its condition trained on, a coverage/quality trade-off, not degradation. Dashed line: chance (0.002).

Two secondary observations close the comparison. **Untrained languages stay well above chance**: the Latin model reaches R@1 0.100 on Chinese, 50× chance, despite never seeing a Chinese caption, which is M-CLIP's pretrained cross-lingual transfer showing through the frozen encoder. **FAR achieves the best FID** (63.924) despite weaker alignment than Latin, consistent with the dissociation of Section 4.8.

**The cosine term restores diversity.** This is the clearest practical result in the comparison. Adding the cosine term to the contrastive objective (*ALL + cosine*) roughly *doubles* diversity, from 0.73–0.83 to 1.459, second only to the English-only model, at essentially unchanged FID (74.479 vs 74.035). This is exactly the mechanism of Section 4.8: contrastive CE is scale-free and constrains only ranking, so nothing in it forces full-amplitude output; the cosine term supplies the missing absolute constraint (raising mean similarity from ∼0.10 to ∼0.55) while CE supplies the per-sample separation cosine alone can never produce. **The two objectives are complementary, and the combined loss is the recommended configuration**: near-best diversity, the most uniform cross-lingual coverage of any condition (spread 0.0066), at a modest cost in peak English retrieval.

CHAPTER 5

# Discussion

## 5.1 What the measurements support

Four claims are supported by the evidence above. (1) In a single-latent motion VAE, sampling, not KL strength, is the binding constraint on reconstruction (Section 4.3), and the first-order argument of Section 4.3.1 explains why. (2) The cosine objective produces no per-sample alignment at any weight, while a contrastive objective produces a real, monotone, cross-lingually structured alignment (Sections 4.6–4.7). (3) Alignment and generation quality dissociate (Section 4.8). (4) Under MSE reconstruction, generation amplitude decays toward a mean pose (Section 4.9), and multilingual conditioning divides a fixed representational budget rather than enriching it (Section 4.11). The one clearly *positive* design recommendation is the combined CE+cosine loss, which restores diversity at no FID cost.

## 5.2 Threats to validity

This section is deliberately blunt; it is what makes the rest credible.

- **Machine-translated captions** confound translation artefacts with genuine typological variation. No human-authored non-English captions exist in this study.
- **Per-language exposure is not matched.** Under $N$ languages, each clip is seen in a given language $\sim 1/N$ as often at a fixed number of gradient steps. Compute is matched; exposure is not. Any multilingual degradation may reflect reduced English exposure rather than multilinguality (Section 4.11).
- **R-precision is English-only by construction**: its text tower is a GloVe BiGRU trained on English. Per-language quality is assessed by FID alone.
- $\lambda_{\mathrm{KL}}$ **was compared against zero, not swept.**
- **Alignment curves had not converged** at the epoch budget used (Figure 4.2, Figure C.1).
- **FID values are** $100\times$ **above published HumanML3D results**; MotionCLIP is a weak generator and the VAE pool is not the official test split. All comparisons here are relative, not absolute.
- **The VAE fine-tune changed two variables at once** (Section 4.4); the deterministic-only control is outstanding.

## 5.3 Toward a sequence of latents

Every failure in Chapter 4 traces to one cause: a single 512-dim latent is the wrong container for a 60-frame, multilingually conditioned motion. The bottleneck cannot absorb sampling noise (Equation 4.1), one text embedding cannot supply trajectory across 60 frames (Section 4.10), and MSE regresses that single vector toward a mean pose (Section 4.9).

The architectural fix is a **sequence of latents**. As Figure 4.1 shows, distributing the clip over $K$ latents with disjoint frame spans makes the noise-damage term block-diagonal and lets one text embedding condition local segments rather than the whole clip. We predict specifically

that **amplitude collapse should not appear under *discrete* latents**, because quantisation prevents regression toward a continuous conditional mean, there is no static pose to average into. The natural test is **MoMask** [6], whose residual-VQ token sequence is exactly this design, conditioned on M-CLIP text. This is the single most valuable next experiment, and it invalidates every checkpoint produced here, which is the honest reason it was left as the forward proposal rather than started.

CHAPTER 6

# Conclusion

The internship set out to answer whether multilingual conditioning helps or harms 3D motion generation once capacity and compute are matched, and whether independently trained per-language projections share a common space. Under a controlled comparison, added languages *divide* a fixed representational budget: they improve coverage of the languages they train on while degrading English monotonically, and they do not supply complementary motion signal (Section 4.11).

Reaching a comparison that could be trusted required correcting the measurement itself. Mean text–motion cosine, the obvious diagnostic, is actively misleading for this model class: it can sit near 1.0 while per-sample alignment is zero (Section 4.6). Replacing it with a matched-minus-mismatched gap, and the cosine *objective* with a contrastive one, produced alignment that grows monotonically and with stable cross-lingual structure, but at a cost in generation, because discrimination and decodability dissociate (Section 4.8). Along the way we established that sampling, not KL strength, breaks a single-latent motion VAE (Section 4.3), recovered a working variational model (Section 4.4), and diagnosed an amplitude collapse driven by MSE reconstruction (Section 4.9). The one positive design result is the combined contrastive-plus-cosine loss, which restores diversity at no FID cost and gives the most uniform cross-lingual coverage of any condition.

What remains open is architectural and specific: move from one latent to a sequence of discrete latents, and re-run the multilingual comparison on MoMask with M-CLIP conditioning. Every measurement in this report points at that experiment. The repository, the experiment index (Appendix B), and the handover notes are organised so that it can be started on day one.

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APPENDIX A

# Implementation details

## A.1 Hardware and training

All experiments ran on a single **RTX 5090** (32 GB), shared. Training at batch 128 runs ∼32 s/epoch. Long runs were detached with `setsid` (not `nohup`+`disown`, which did not survive SSH disconnects on this machine). Default training configuration: batch 128, learning rate $10^{-4}$, 60 frames, latent 512, 8 layers, GELU, $\lambda_{\text{rc}} = 100$, $\lambda_{\text{vel}} = 100$, $\lambda_{\text{KL}} = 10^{-3}$.

## A.2 Reusable tools written during this work

- `eval_latent_alignment.py`, matched/mismatched gap and retrieval, 7 languages, per checkpoint.
- `plot_training_curves.py`, parses `training.log`, produces convergence figures.
- `translate_nllb.py`, NLLB caption translation into 6 target languages.
- Patches `patch_vae.py`, `patch_anneal.py`, `patch_mclip.py`, `patch_ml.py`, `patch_perlang.py`, VAE terms, KL annealing, M-CLIP substitution, multilingual sampling, per-language bank.
- `eval_all.sh`, evaluates every `ce150-*` condition (alignment + generation), the source of Section 4.11.

## A.3 Known bugs and traps

- **`reparameterize()` couples $\lambda_{\text{KL}}$ to sampling.** Sampling activates as soon as $\lambda_{\text{KL}} \neq 0$, so every KL-annealing run trained with full-magnitude noise from epoch 1 while only the penalty was ramped. This is why annealing appeared to fail (Section 4.4).
- **Evaluation scripts hardcode checkpoints.** `eval_guo_vae_compare_beta05.py` and `gen_hml.py` now accept `--ckpt`; `eval_noise_val100.py` still does *not* and ignores it.
- **`gen_hml.py` assumed plain CLIP** (called `clip.tokenize`); patched to `encode_text_raw` with a `--text_encoder` flag that *must* match how the checkpoint was trained, or the output is silently meaningless.
- **`logit_scale` is missing from both custom text encoders**; added as a learnable (frozen) parameter to `mclip_wrapper.py` and `per_lang_encoder.py`, without which the CE path crashes.
- **Training losses are not evidence.** CE loss stayed near ln(batch) while alignment improved 100×, and reconstruction looked healthy while motion collapsed. Always verify against an evaluation metric.

APPENDIX B

# Experiment index

Every run that produced a checkpoint, the command that produced it, its key configuration, what it tested, and its headline result. Section numbers refer to Chapter 4. Only the final checkpoint of each run is kept, except where the epoch progression is itself the result.

| Run | Key config | Tested | Headline / § |
|---|---|---|---|
| `ml-en` | M-CLIP, en | baseline | FID 18.013 · §4.1 |
| `ml-latin` | M-CLIP, en/fr/es/pt | multilingual baseline | FID **16.140**, best · §4.1 |
| `ml-far` | M-CLIP, en/ru/zh/ar | multilingual baseline | FID 18.046 · §4.1 |
| `ml-all` | M-CLIP, all 7 | multilingual baseline | FID 17.829 · §4.1 |
| `humanml3d-263`, `humanml3d-mclip`, `paper-model`, `my-paper-model` | original MotionCLIP | reproduction | baselines before modification (not cited) |
| `perlang-latin` | 4 frozen BERTs, cosine | per-lang encoders | FID 24.55, R@1 0.079 · §4.2 (uninformative: cosine) |
| `vae-en-kl1e-5` | $\lambda_{\text{KL}} = 10^{-5}$, sampling | KL strength | rc 0.524 · §4.3 |
| `vae-en-kl1e-3` | $\lambda_{\text{KL}} = 10^{-3}$, sampling | KL strength | rc 0.526 · §4.3 |
| `vae-en-actor` | cosine removed | ACTOR-style | rc 0.524 · §4.3 |
| `vae-en-anneal` | anneal 30 ep | KL schedule | rc 0.525 · §4.3 |
| `vae-en-mu-kl0` | recon from $\mu$, $\lambda_{\text{KL}}$=0 | det. recon | FID 97.633 · §4.4 |
| `vae-en-mu-kl1e-3` | recon from $\mu$ | det. recon + KL | FID **55.864** · §4.4 |
| `vae-en-mu-kl1e-3-anneal30/5` | anneal 30 / 5 ep | KL schedule | FID 105.7 / 100.9 · §4.4 |
| `vae-en-mu-beta*-ft`, `*-distill`, `*-decoderonly`, `*-musem` | $\beta$ fine-tunes | noise robustness | +25.5% rc, FID 57.3→86.8 · §4.5 (closed) |
| `vae-multilingual-mclip` | 7 langs, cosine | alignment | gap 0.0002 · §4.6 |
| `vae-ml-cos100` | cosine ×100 | alignment | gap 0.0007 (worse) · §4.6 |
| `vae-ml-ce` | CE, batch 64 | alignment | gap 0.0206 · §4.6 |
| `vae-ml-ce-s100` | CE, logit_scale 100, b128 | alignment | gap 0.0186 · §4.7 |
| `vae-ml-ce-s100-e100` | CE, 5 checkpoints | gap vs epoch | monotone, Fig. 4.2 · §4.7 |
| `vae-ml-ce-s100-e300` | CE, 6 checkpoints | gap, generation | FID 72.902, R@10 0.340 · §4.7,4.8 |
| `vae-ml-sampled-ft` | det.-then-sampled, 330 ep | working VAE | rc **0.086**, amp 0.0044 · §4.4,4.9 |
| `vel10-test` | $\lambda_{\text{vel}} = 10$ | velocity weight | amp 0.0145 · §4.9 |
| `ce150-combined` | CE+cosine, 3 ckpts | amplitude decay | amp 0.0108 · §4.9,4.11 |
| `ce150-all` | CE, all 7, 3 ckpts | amplitude decay | amp 0.0063 · §4.9 |
| `wk-rc20` | $\lambda_{\text{rc}} = 20$ | recon weight | amp 0.0078 · §4.10 |
| `wk-textrc` | text_rc on, $\lambda_{\text{rc}}$=100 | text-cond. recon | conditioning ×6, amp 0.0042 · §4.10 |
| `wk-textrc-rc20` | text_rc on, $\lambda_{\text{rc}}$=20 | text-cond. recon | conditioning ×7, amp 0.0030 · §4.10 |
| `wk-perlang`, `wk-perlang-comb` | per-lang bank, CE | encoder choice | amp 0.017–0.019, reverses §4.2 · §4.10 |
| `ce150-en/latin/far/all` | CE, 150 ep, language set | controlled comparison | Table 4.14 · §4.11 |

**Reproduction.** Each command reproduces the corresponding block of results; `<run>` and the checkpoint epoch are filled from the table above.

```
# alignment gap + retrieval, per language
python eval_latent_alignment.py
```

```
    --ckpt exps/<run>/checkpoint_XXXX.pth.tar
    --tag <name> --n 500 --text_encoder mclip

# generation quality (evaluator B)
python eval_guo_vae_compare_beta05.py
    --model original --replications 20
    --device 0 --ckpt exps/<run>/checkpoint_XXXX.pth.tar

# the controlled language comparison (§4.11)
bash eval_all.sh 2>&1 | tee results_all.log
```

**Open threads.** (1) Add velocity supervision to the text-conditioned path (Section 4.10). (2) Run 10 further *deterministic* epochs from `vae-ml-ce-s100-e300` to control the VAE finetune (Section 4.4). (3) Sweep $\lambda_{\mathrm{KL}}$. (4) Human-authored non-English captions. (5) The sequence-of-latents architecture (Section 5.3).

APPENDIX C

# Additional tables and figures

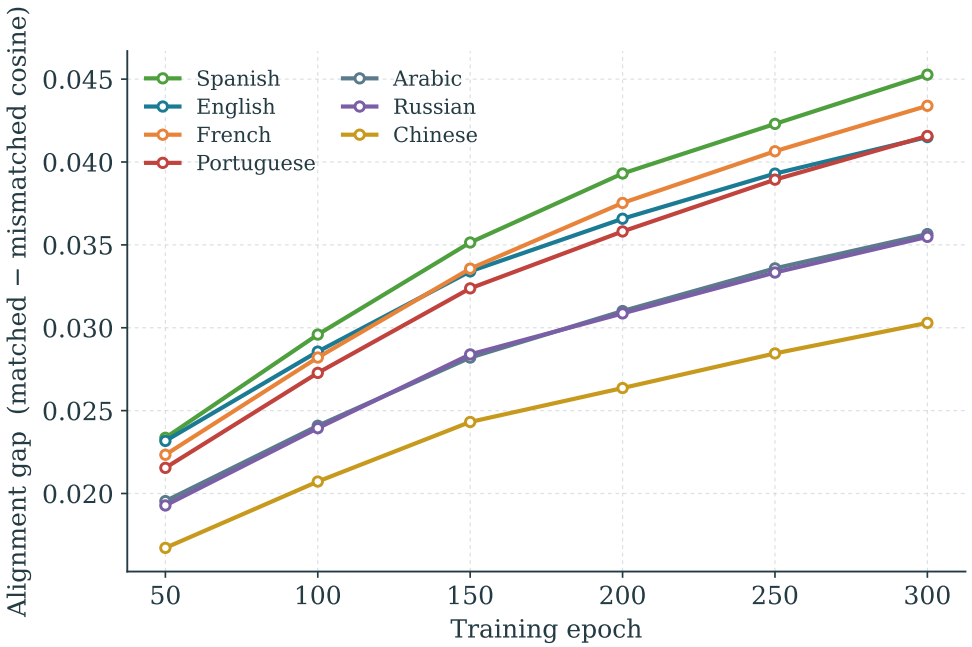


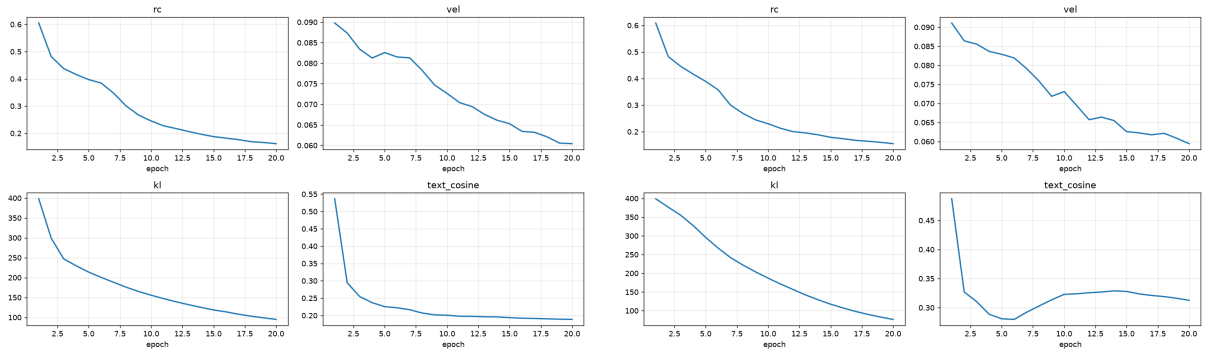

**Figure C.1.** Alignment gap versus epoch over the long 300-epoch run, one line per language. No plateau at 300 epochs; the cross-lingual ordering is the same as in Figure 4.2. This is the evidence for the "curves had not converged" limitation in Section 5.2.

**Table C.1.** Per-language alignment gap by training condition (500 clips, parallel captions), companion to Table 4.15.

| condition | en | fr | es | pt | ru | zh | ar |
|---|---|---|---|---|---|---|---|
| EN | 0.0515 | 0.0265 | 0.0276 | 0.0249 | 0.0215 | 0.0167 | 0.0182 |
| Latin | 0.0372 | 0.0403 | 0.0420 | 0.0388 | 0.0232 | 0.0178 | 0.0206 |
| FAR | 0.0336 | 0.0248 | 0.0262 | 0.0232 | 0.0302 | 0.0268 | 0.0316 |
| ALL | 0.0334 | 0.0336 | 0.0351 | 0.0324 | 0.0284 | 0.0243 | 0.0282 |

**Table C.2.** Combined loss (CE + cosine), per-language matched/mismatched cosine and gap. The cosine term raises absolute similarity to ∼0.55 while preserving a real gap.

| | en | fr | es | pt | ru | zh | ar |
|---|---|---|---|---|---|---|---|
| matched | 0.5460 | 0.5450 | 0.5460 | 0.5461 | 0.5426 | 0.5395 | 0.5430 |
| mismatched | 0.5171 | 0.5157 | 0.5153 | 0.5178 | 0.5181 | 0.5185 | 0.5187 |
| **gap** | 0.0289 | 0.0293 | 0.0307 | 0.0282 | 0.0245 | 0.0209 | 0.0243 |




**(a)** `vae-en-mu-kl1e-3` (English). **(b)** `vae-multilingual-mclip` (7 languages).

**Figure C.2.** Training curves (rc, vel, KL, text_cosine) for the English and multilingual VAEs, produced by `plot_training_curves.py`.

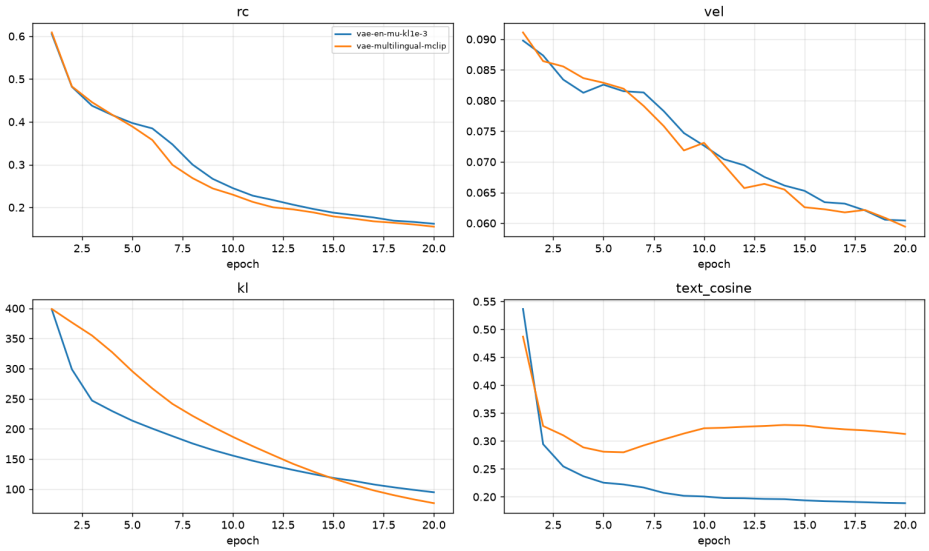



**Figure C.3.** Direct overlay of the English and multilingual VAE training curves. Reconstruction, velocity, and KL track closely; the two diverge only in `text_cosine`, where the multilingual model settles higher, a difference Section 4.6 shows to be uninformative about per-sample alignment.

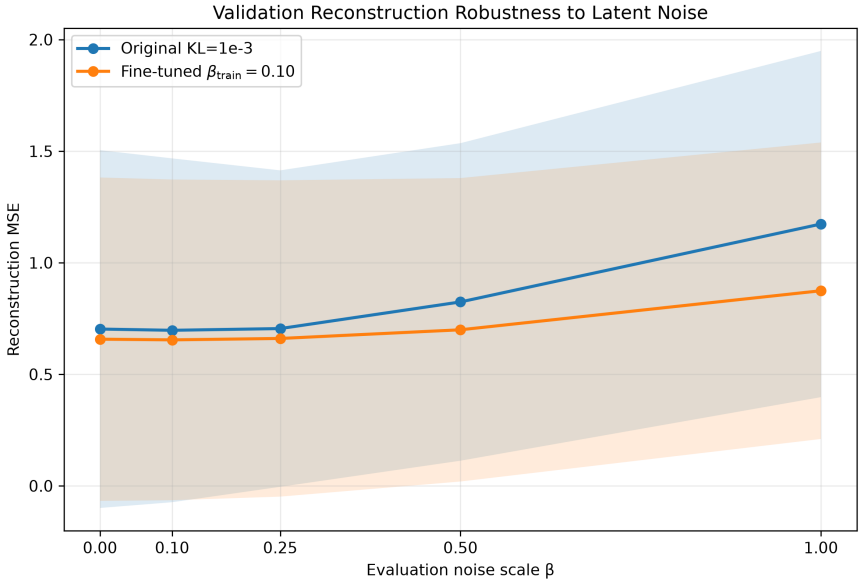



**Figure C.4.** Validation reconstruction under injected noise $z = \mu + \beta\varepsilon$ for the noise-robustness branch of Section 4.5. The $\beta = 0.10$ fine-tune improves reconstruction under noise (the reported +25.5%) but degrades generation (Table 4.6).